VAE vs. DDPM

for Image Generation on Fashion-MNIST

A controlled comparison of distribution quality, diversity, and computational efficiency

RESEARCH QUESTION

How much generation quality does iterative diffusion gain over latent-variable decoding — and what computational cost accompanies it?

CONTRIBUTION

Convolutional VAE vs. 100-step DDPM · 5,000 samples/model · shared Fashion-MNIST feature evaluator

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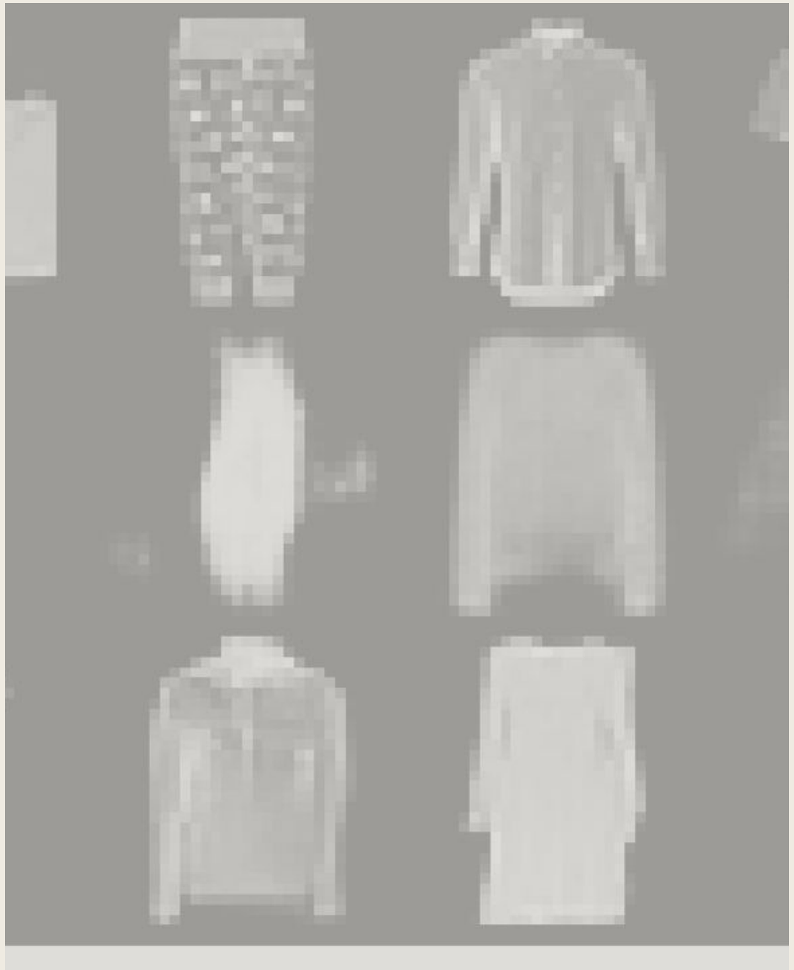
Course / Institution / Date

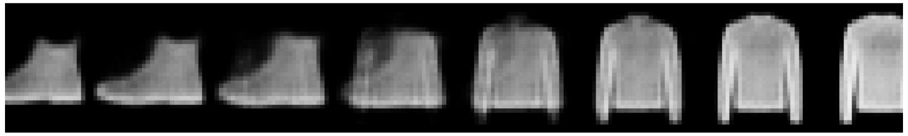


Kingma & Welling, 2013; Ho et al., 2020; Xiao et al., 2017

Same evaluation pipeline; different generation mechanisms.

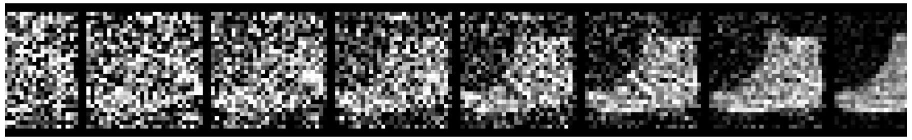
Controlled setup: Fashion-MNIST 60k/10k · 28×28 grayscale · 5,000 generated samples/model · CNN evaluator 91.34% accuracy · seed 42

VAE Latent-variable decoding



Encoder q(z|x) learns a 32-D latent space
Generate: sample z ~ N(0,I), then one decoder pass
1.70M parameters · 25 epochs

DDPM Iterative reverse denoising



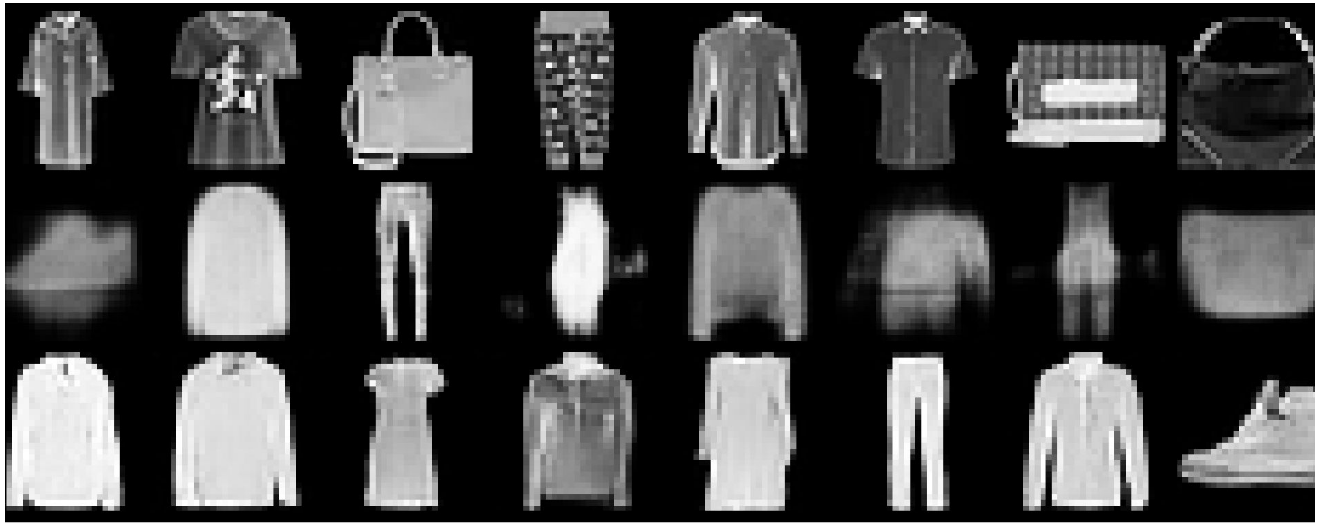
Noise-prediction U-Net with cosine schedule
Generate: 100 sequential reverse-denoising steps
5.22M parameters · 35 epochs

Quality: FID/KID + confidence **Diversity: coverage + entropy** **Cost: time + latency + memory**

Qualification: controlled system comparison, not a capacity-matched causal ablation.

Metrics use shared Fashion-MNIST classifier features; architectures are not parameter matched.

DDPM samples are closer to the real feature distribution and more balanced.



Real → VAE → DDPM comparison grid

51.1 → 7.7

FID ↓

features closer to real

1.87 → 0.33

KID ↓

same direction

41% → 66%

conf. ≥ 0.8 ↑

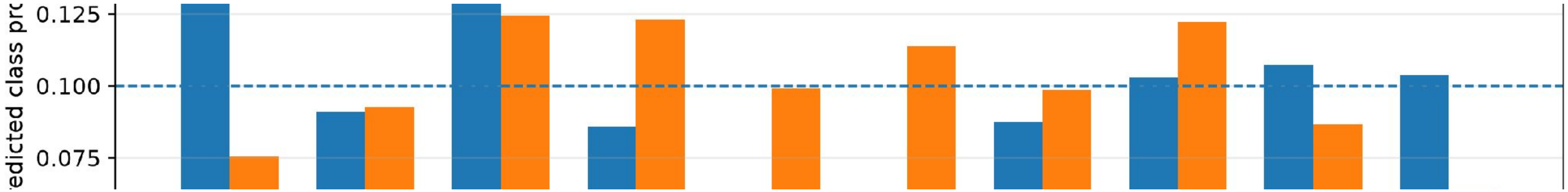
more recognizable

0.931 → 0.991

entropy ↑

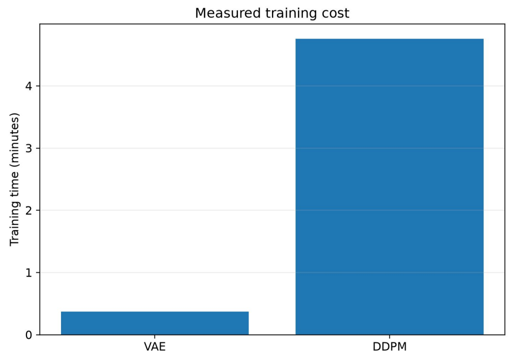
more balanced

Coverage alone hides imbalance: both reach all 10 classes, but the VAE overproduces several categories.

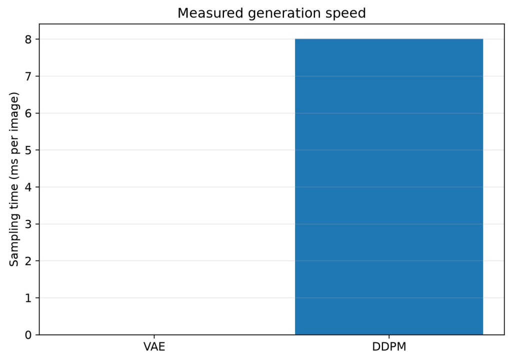


FID/KID use Fashion-MNIST classifier features, not ImageNet Inception features.

DDPM quality comes from substantially greater sequential computation.



Measured training time



Batched generation latency

Resource multiplier: DDPM / VAE

12.8× training time

~1,068× sampling latency

11.2× peak GPU memory

3.1× parameters

Why latency explodes: parameter count rises only 3.1×, but DDPM generation evaluates the U-Net 100 times. The VAE uses one decoder pass.

VAE strength: fast, compact, smooth latent interpolation
Limitation: blurrier samples and weaker feature match

DDPM strength: stronger FID/KID, confidence, sharpness, balance
Limitation: sequential sampling and higher memory cost

Measured on one RTX 5070 Ti run with mixed precision and large batches; absolute latency is hardware- and implementation-dependent.

Neither model dominates: diffusion buys quality; VAE buys speed.

Choose DDPM when the priority is...

- Distribution match and recognizability
- Balanced predicted class coverage
- Fidelity matters more than inference cost

Evidence: lower FID/KID, higher confidence, entropy 0.991 vs. 0.931

Choose VAE when the priority is...

- Low-latency generation
- Lower memory/training budget
- Direct 32-D latent interpolation

Evidence: ~1,068× lower ms/image; 12.8× less training time; 11.2× less peak allocation

Validity limits

one seed · unequal capacity · classifier-feature metrics · simple 28×28 grayscale data

Next experiments

3+ seeds · matched parameter/compute budget · 10/25/50/100 DDPM-step Pareto curve · per-class diversity

Takeaway: in this controlled implementation, DDPM is the quality-oriented choice; VAE is the efficiency-oriented choice.

Conclusions are limited to this controlled Fashion-MNIST implementation.